

# STA 2101: Statistics & Probability

## Lecture 04: Data Distribution, Charts

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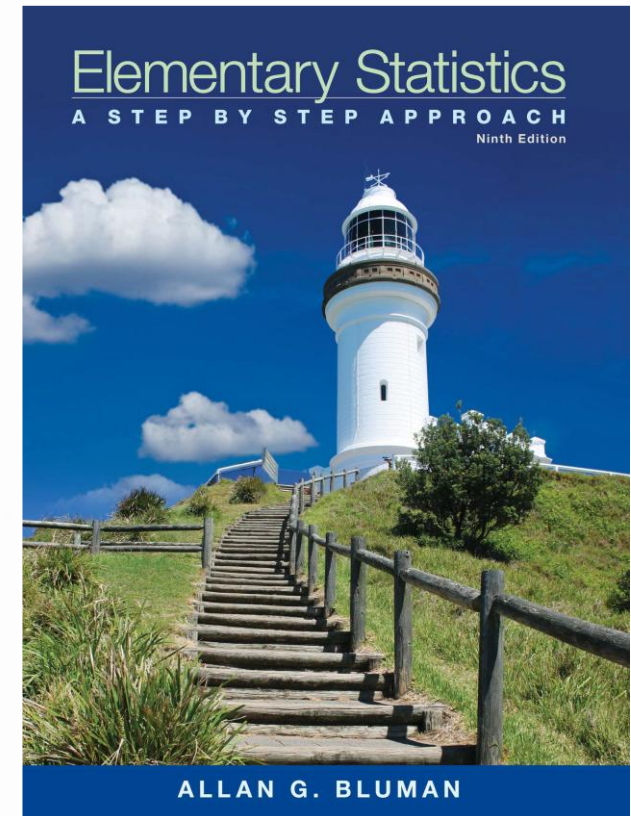
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Sec 5 - ayvzsvgm

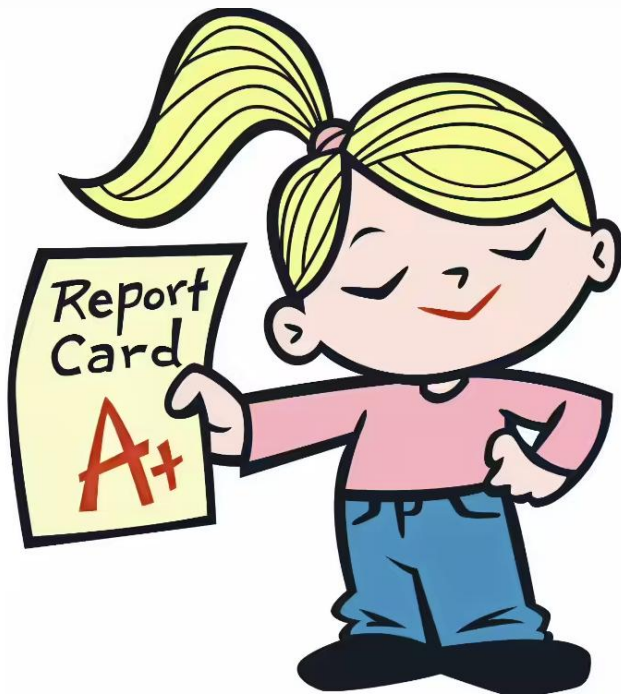
Spring 2026

# Reference Book

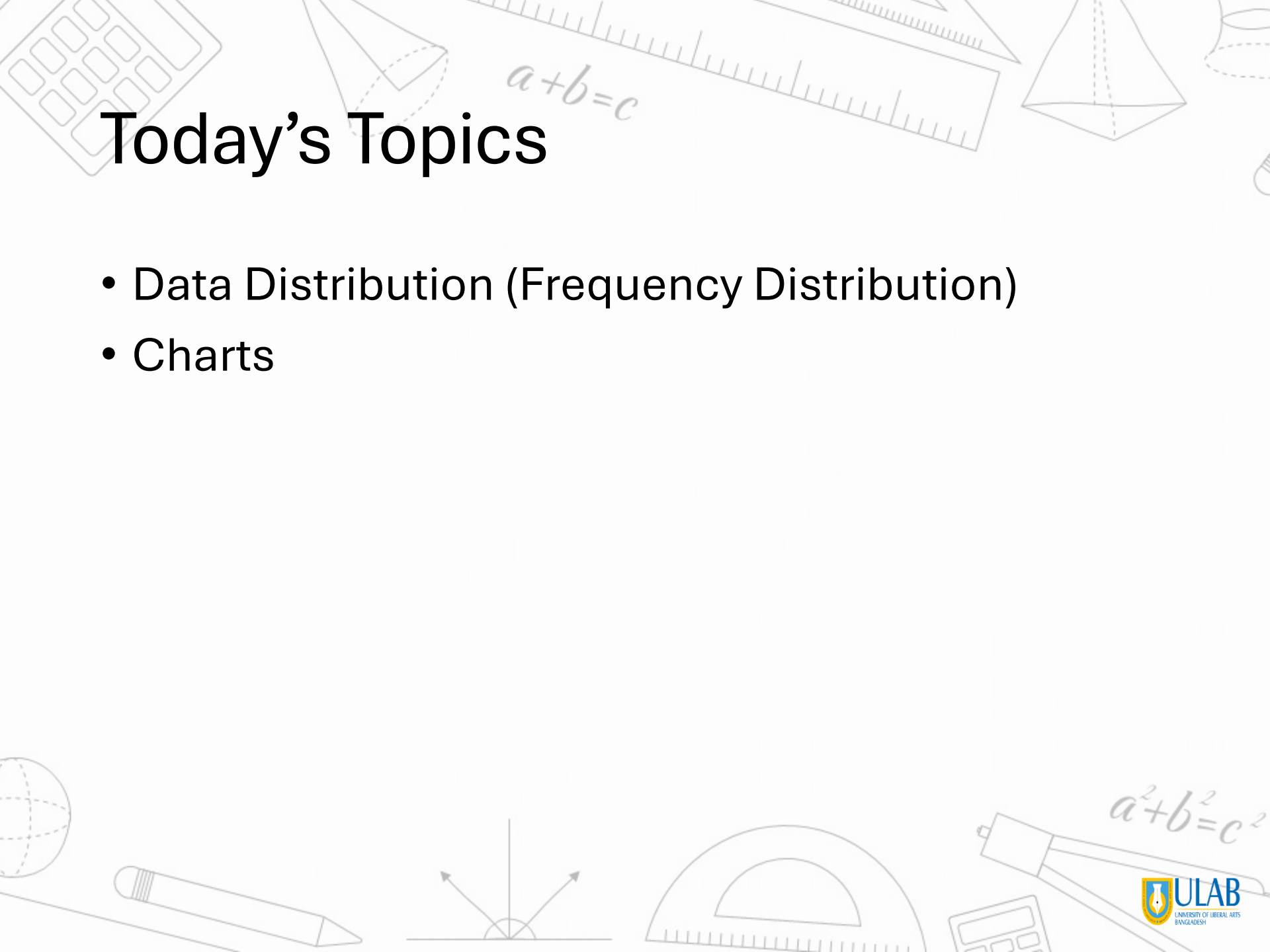
- **Text Book(s):**
- Elementary Statistics – Allan G. Bluman
- Introduction to Probability and Statistics (14th Edition) By William Mendenhall, Robert J. Beaver, Barbara M. Beaver
- **Reference Material/Book(s):**
- S Introduction to probability and statistics for engineers and scientists, Sheldon M Ross



# Mark Distribution

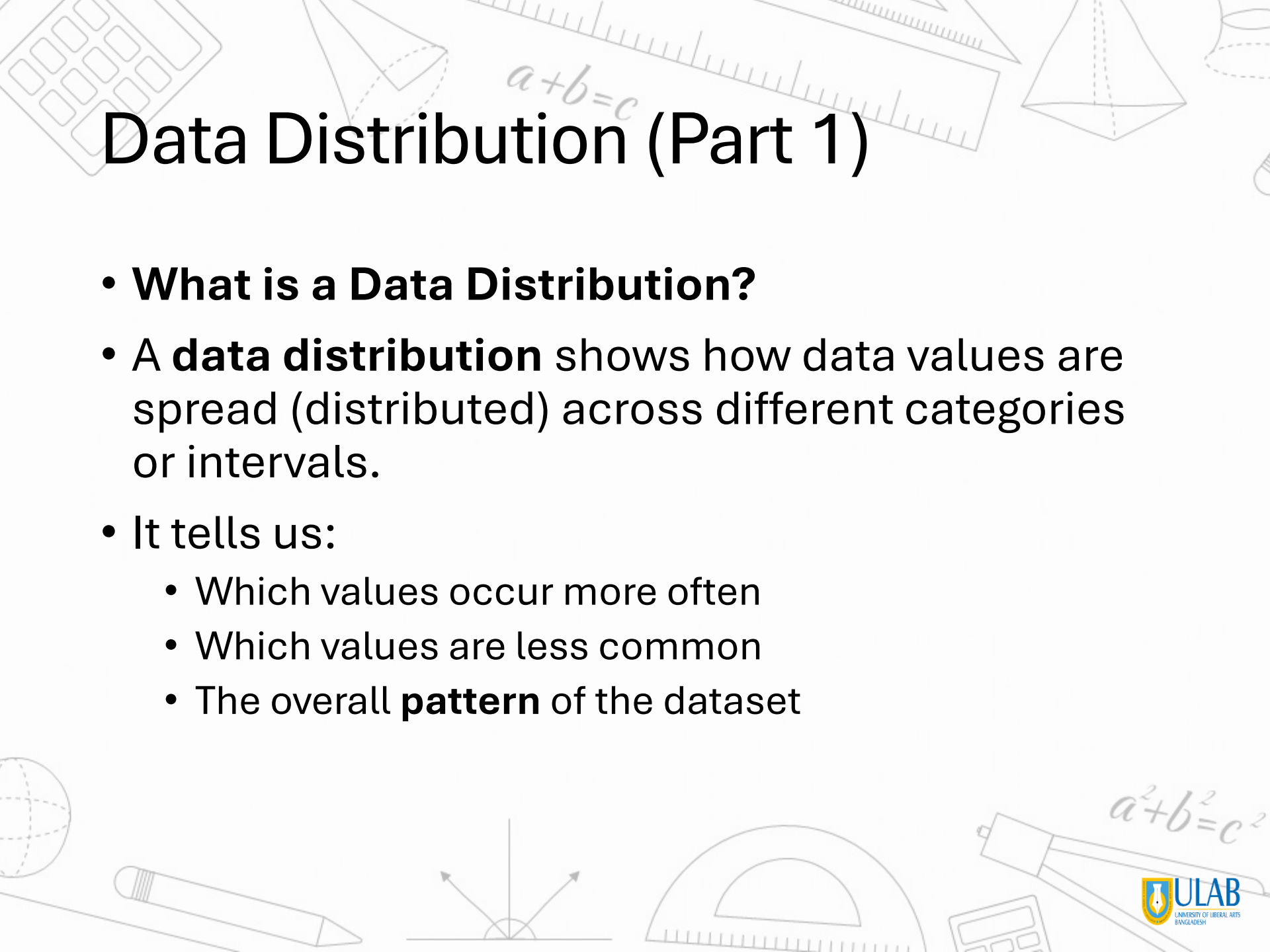


| Assessments                        | %   |
|------------------------------------|-----|
| Mid Term Examination               | 25  |
| Final Examination                  | 40  |
| Attendance and Class Participation | 5   |
| CT                                 | 5   |
| Project                            | 25  |
| Total                              | 100 |

The background of the slide is decorated with various mathematical and geometric illustrations. At the top, there is a ruler, a cone, a pyramid, and a cylinder. In the upper center, the equation  $a+b=c$  is written. On the left, a portion of a calculator is visible. At the bottom, there is a globe, a pencil, a diagram of three angles sharing a common vertex, a semi-circular protractor, a compass, and another calculator. The equation  $a^2+b^2=c^2$  is written near the compass. The University of Liberal Arts Bangladesh (ULAB) logo is in the bottom right corner.

# Today's Topics

- Data Distribution (Frequency Distribution)
- Charts

The background of the slide is decorated with various mathematical and scientific icons. At the top, there's a ruler, a pencil, and a cone. In the center, the equation  $a+b=c$  is written. To the right, there's a pyramid and a cylinder. At the bottom, there's a globe, a pencil, a protractor, a compass, and the equation  $a^2+b^2=c^2$ . The ULAB logo is in the bottom right corner.

# Data Distribution (Part 1)

- **What is a Data Distribution?**
- A **data distribution** shows how data values are spread (distributed) across different categories or intervals.
- It tells us:
  - Which values occur more often
  - Which values are less common
  - The overall **pattern** of the dataset

# Types of Distributions

- **Frequency Distribution**

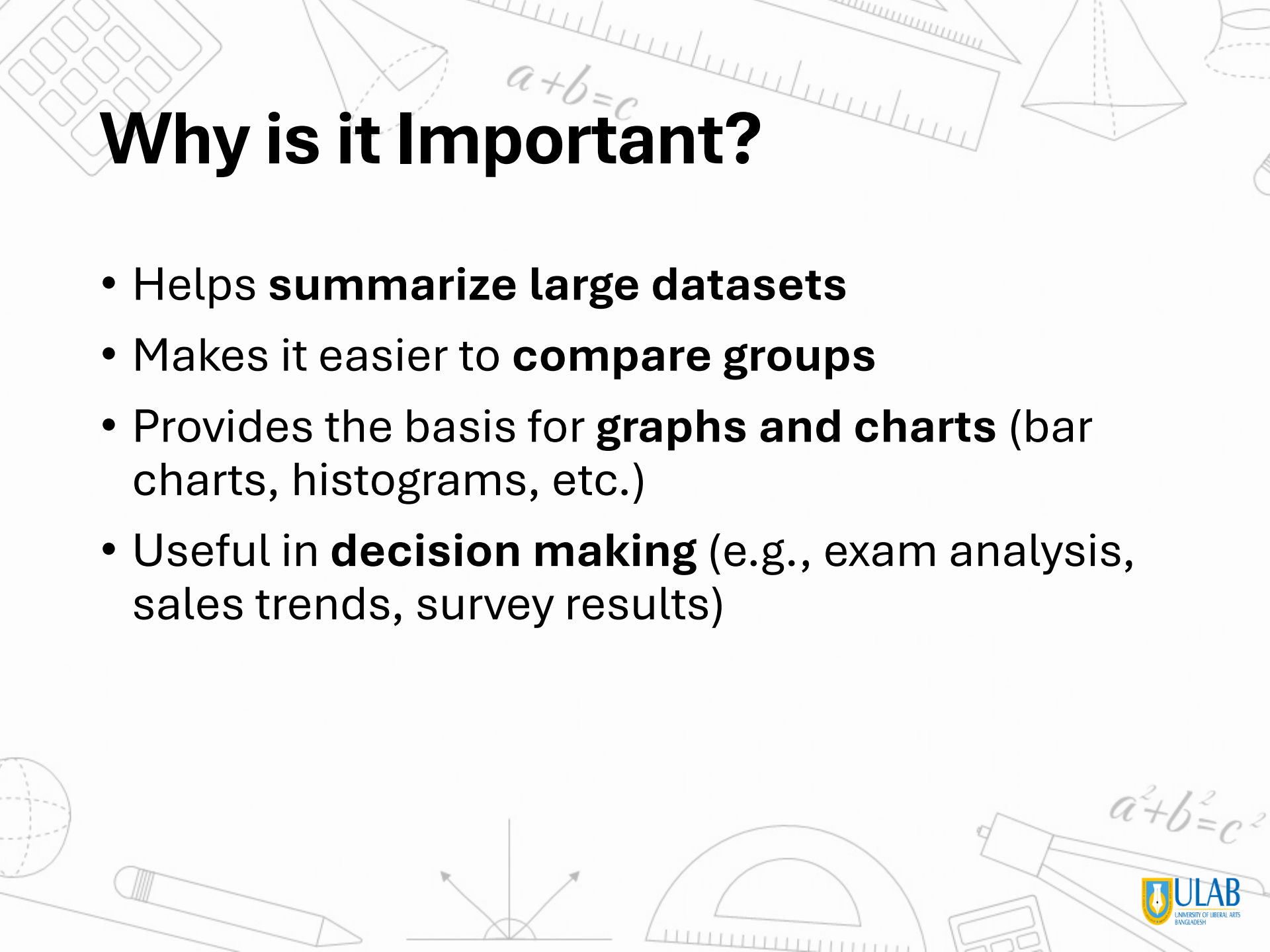
- Shows how many times each value/category occurs.
- Example: Number of students getting grades A, B, C, D.

- **Relative Frequency Distribution**

- Shows the **proportion or percentage** of values in each category.
- Example: 25% students got A, 40% got B, etc.

- **Cumulative Frequency Distribution**

- Shows the **running total** of frequencies.
- Example: How many students scored  $\leq 70$  marks.

The background of the slide is decorated with various mathematical and scientific icons. At the top, there's a ruler, a pencil, and a cone. In the center, the equation  $a+b=c$  is written. On the right, there's a pyramid and a cylinder. At the bottom, there's a globe, a pencil, a protractor, a compass, and the equation  $a^2+b^2=c^2$ . The main title "Why is it Important?" is prominently displayed in the upper left area.

# Why is it Important?

- Helps **summarize large datasets**
- Makes it easier to **compare groups**
- Provides the basis for **graphs and charts** (bar charts, histograms, etc.)
- Useful in **decision making** (e.g., exam analysis, sales trends, survey results)



# Types of Representation

- Table
- Graphs

Cumulative frequency table of the ages of survey participants

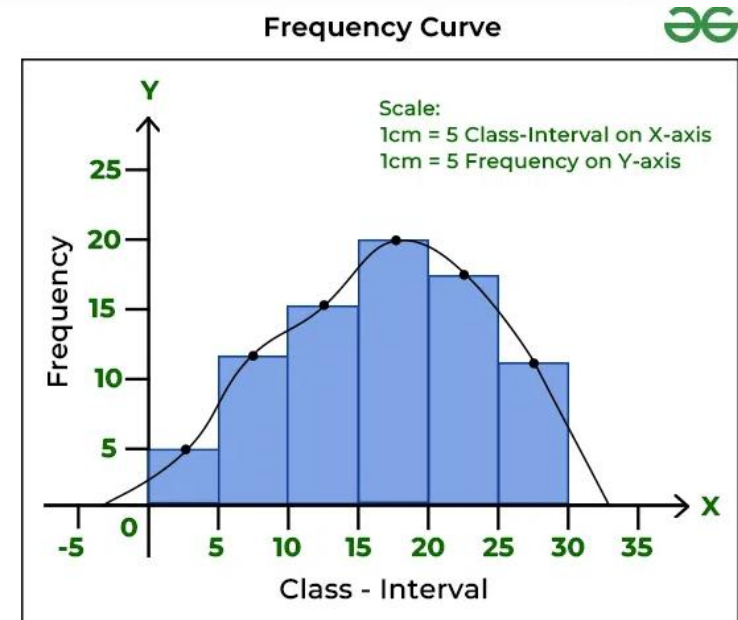
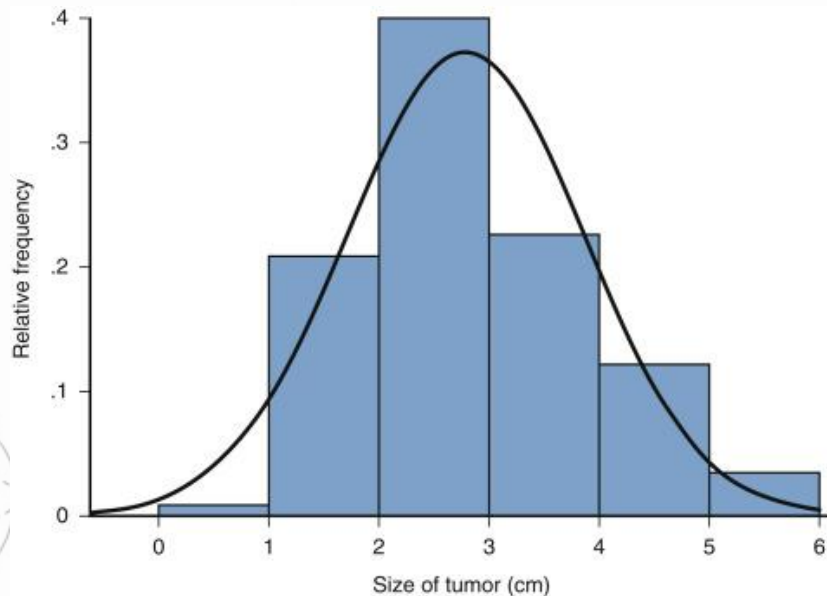
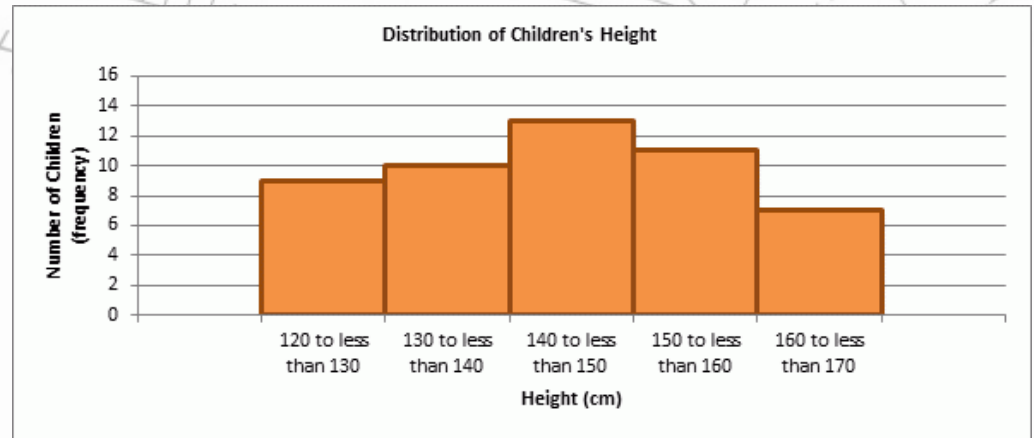
| Age, $a$ (years) | Frequency | Cumulative frequency | Cumulative relative frequency |
|------------------|-----------|----------------------|-------------------------------|
| $19 \leq a < 29$ | 4         | 4                    | $4 / 20 = .2$                 |
| $29 \leq a < 39$ | 9         | $9 + 4 = 13$         | .65                           |
| $39 \leq a < 49$ | 3         | $9 + 4 + 3 = 16$     | .8                            |
| $49 \leq a < 59$ | 3         | 19                   | .95                           |
| $59 \leq a < 69$ | 1         | 20                   | 1                             |

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# Types of Representation

- Table
- Graphs



# Graphs and Charts

- **What are Graphs and Charts?**
- Visual representations of data.
- Make it easier to **see patterns, compare categories, and interpret results.**
- Turn large datasets into **simple, understandable visuals.**

# Common Types of Graphs/Charts

- **Bar Chart**

- Uses rectangular bars to show frequencies or proportions.
- Best for **categorical data** (e.g., favorite fruits, survey responses).

- **Pie Chart**

- Circular chart divided into slices to represent percentages.
- Best for showing **parts of a whole** (e.g., market share, budget distribution).

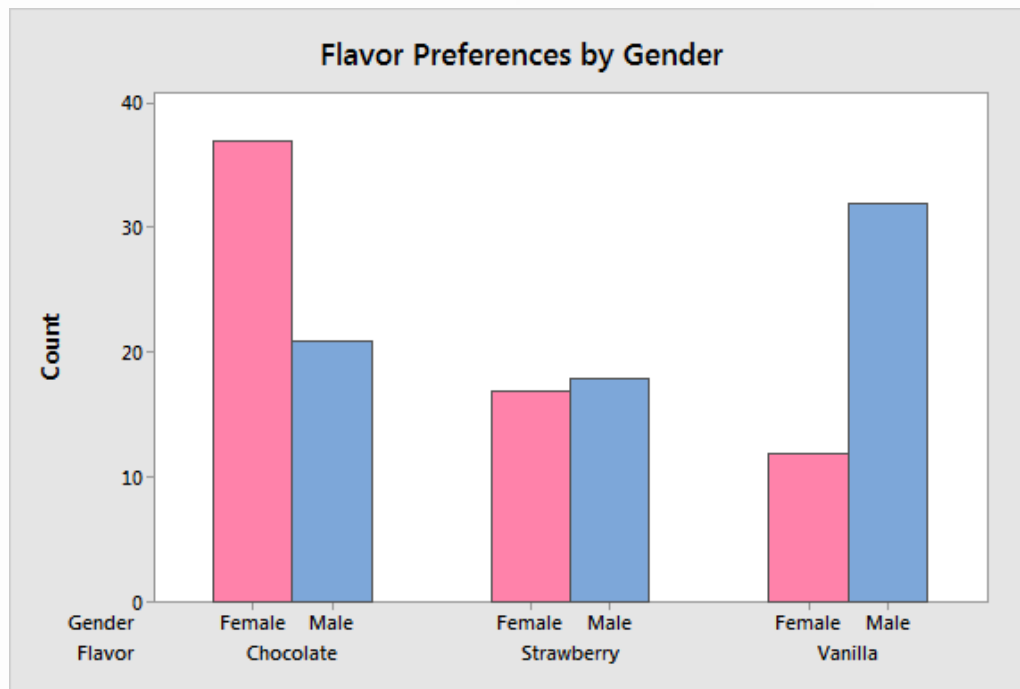
- **Histogram**

- Similar to bar chart but for **continuous data**.
- Shows distribution of data across intervals (e.g., student exam scores).

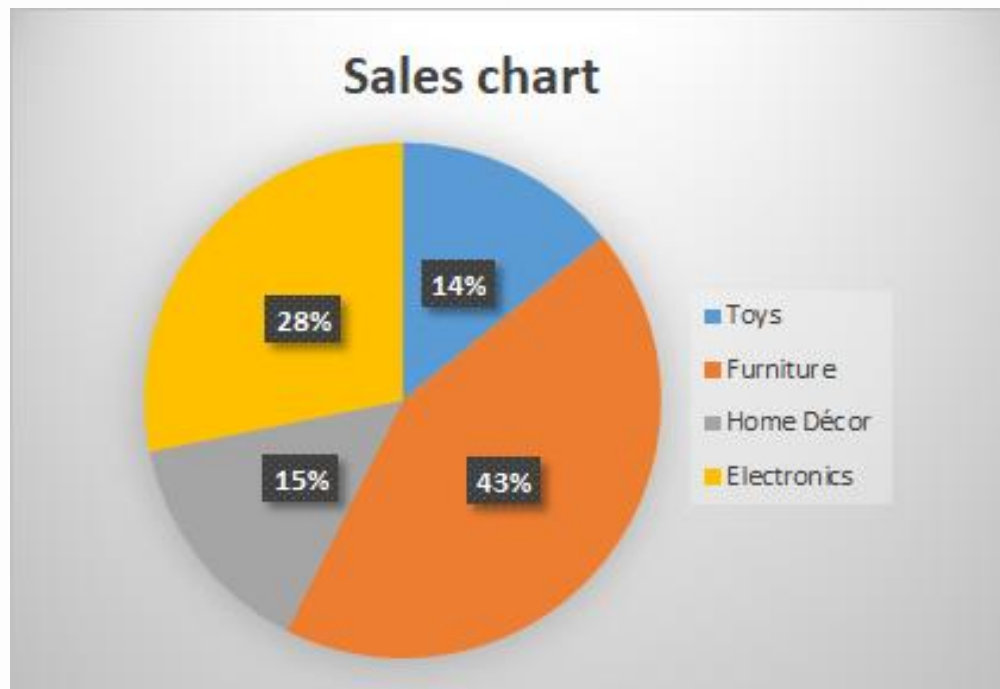
- **Line Graph**

- Shows trends over time with points connected by lines.
- Best for **time series data** (e.g., sales per month, temperature changes).

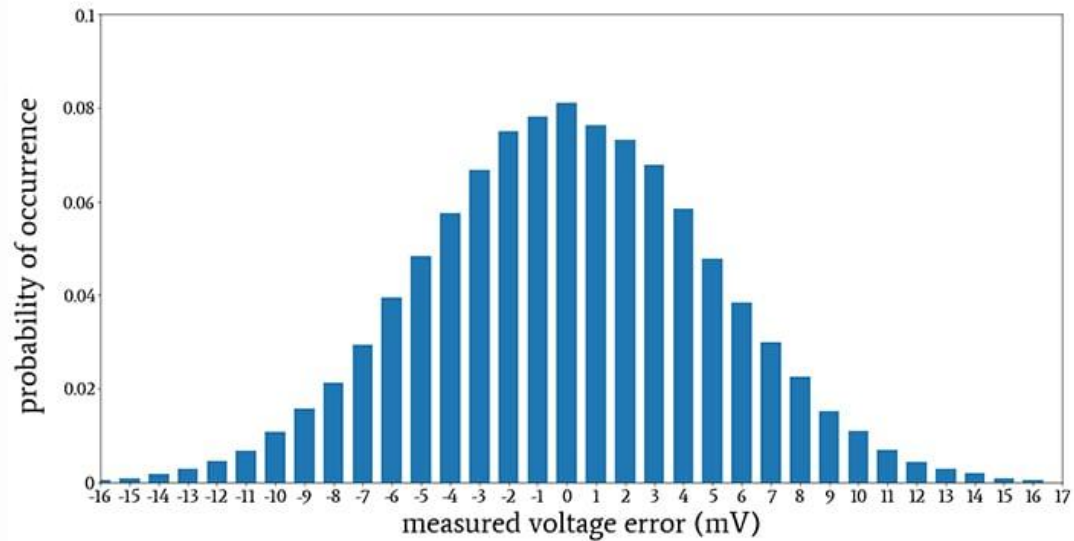
# Bar Chart



# Pie Chart



# Histogram



# Types of Histogram

## Types of Histograms (by Shape of Distribution)

### 1. Uniform Distribution

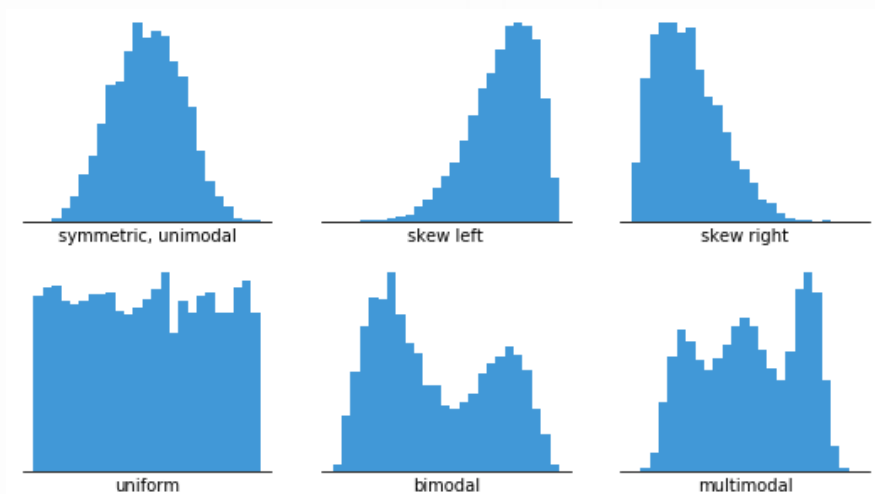
- All bars are roughly the same height.
- Data values are spread out evenly.
- **Example:** Rolling a fair die many times → each number (1–6) occurs equally often.

### 2. Symmetric (Bell-Shaped / Normal Distribution)

- The histogram is **balanced** around the center.
- Bars rise to a peak in the middle and taper off on both sides.
- **Example:** Heights of people, exam scores (in large groups).

### 3. Skewed Right (Positively Skewed)

- Tail (longer side) extends to the **right**.
- Most values are on the lower side, but a few large values pull the tail.
- **Example:** Income distribution (many earn average, few earn extremely high).





# Types of Histogram

## 4. Skewed Left (Negatively Skewed)

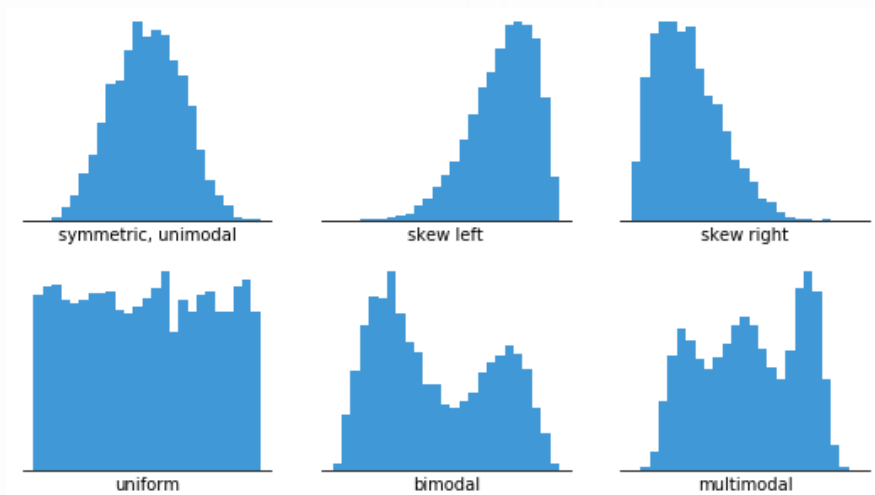
- Tail (longer side) extends to the **left**.
- Most values are on the higher side, but a few very small values pull the tail.
- **Example:** Age at retirement (most retire around 60–65, few retire much earlier).

## 5. Bimodal Distribution

- Two distinct peaks (modes).
- Suggests data comes from **two different groups**.
- **Example:** Test scores of a mixed class (strong group + weak group).

## 6. Multimodal Distribution

- More than two peaks.
- Indicates multiple subgroups in the data.
- **Example:** Daily traffic flow (morning, noon, evening peaks).



# From the book

## 2-2 Histograms, Frequency Polygons, and Ogives

### The Histogram

The **histogram** is a graph that displays the data by using contiguous vertical bars (unless the frequency of a class is 0) of various heights to represent the frequencies of the classes.

#### EXAMPLE 2-4 Record High Temperatures

Construct a histogram to represent the data shown for the record high temperatures for each of the 50 states (see Example 2-2).

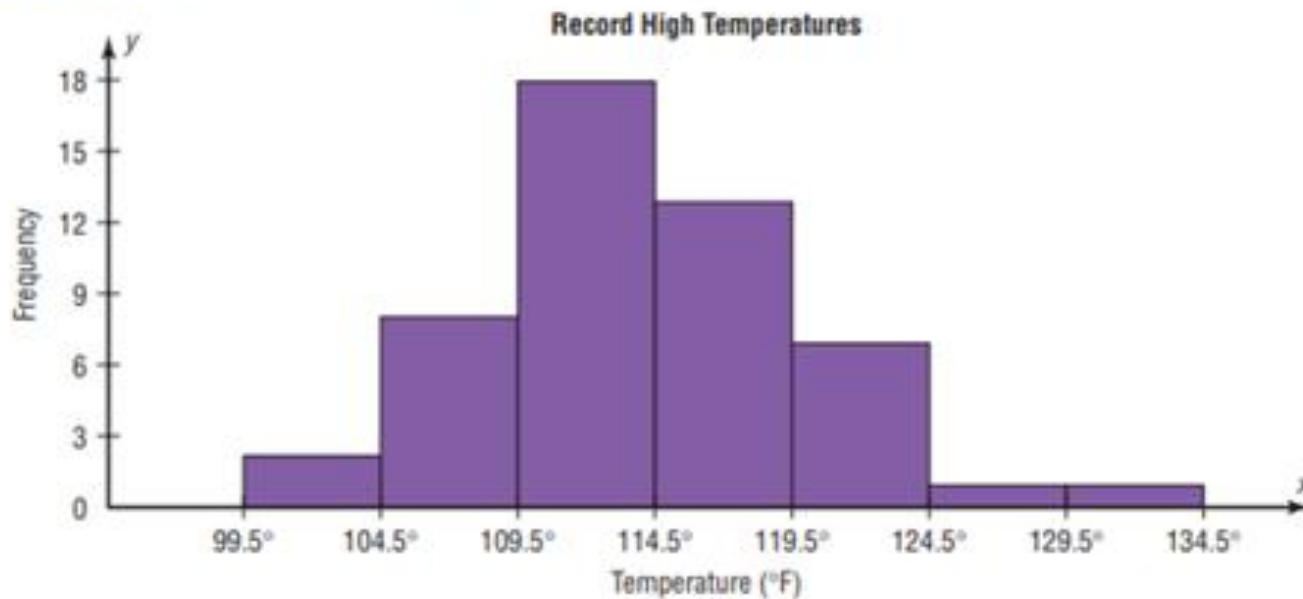
| Class boundaries | Frequency |
|------------------|-----------|
| 99.5–104.5       | 2         |
| 104.5–109.5      | 8         |
| 109.5–114.5      | 18        |
| 114.5–119.5      | 13        |
| 119.5–124.5      | 7         |
| 124.5–129.5      | 1         |
| 129.5–134.5      | 1         |

#### SOLUTION

- Step 1** Draw and label the  $x$  and  $y$  axes. The  $x$  axis is always the horizontal axis, and the  $y$  axis is always the vertical axis.
- Step 2** Represent the frequency on the  $y$  axis and the class boundaries on the  $x$  axis.
- Step 3** Using the frequencies as the heights, draw vertical bars for each class. See Figure 2-1.

# From the book

**FIGURE 2-1** Histogram for Example 2-4



As the histogram shows, the class with the greatest number of data values (18) is 109.5–114.5, followed by 13 for 114.5–119.5. The graph also has one peak with the data clustering around it.

# From the book

## $a+b=$ The Frequency Polygon

Another way to represent the same data set is by using a frequency polygon.

The **frequency polygon** is a graph that displays the data by using lines that connect points plotted for the frequencies at the midpoints of the classes. The frequencies are represented by the heights of the points.

### EXAMPLE 2-5 Record High Temperatures

Using the frequency distribution given in Example 2-4, construct a frequency polygon.

#### SOLUTION

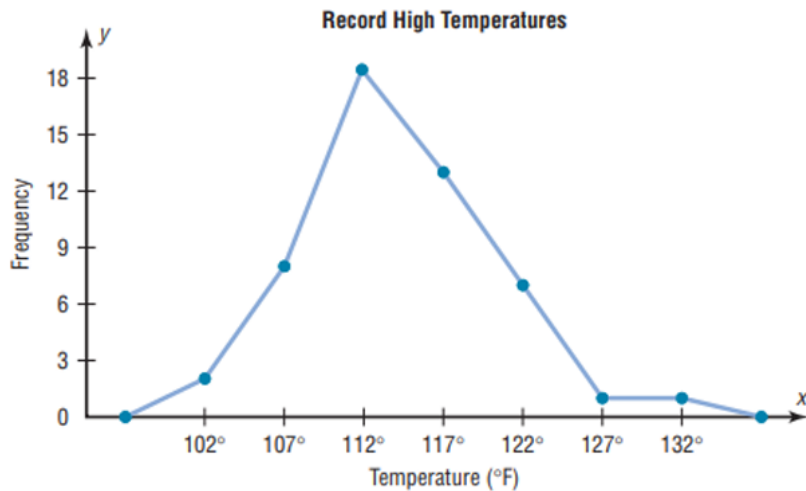
**Step 1** Find the midpoints of each class. Recall that midpoints are found by adding the upper and lower boundaries and dividing by 2:

$$\frac{99.5 + 104.5}{2} = 102 \quad \frac{104.5 + 109.5}{2} = 107$$

and so on. The midpoints are

| Class boundaries | Midpoints | Frequency |
|------------------|-----------|-----------|
| 99.5–104.5       | 102       | 2         |
| 104.5–109.5      | 107       | 8         |
| 109.5–114.5      | 112       | 18        |
| 114.5–119.5      | 117       | 13        |
| 119.5–124.5      | 122       | 7         |
| 124.5–129.5      | 127       | 1         |
| 129.5–134.5      | 132       | 1         |

- Step 2** Draw the  $x$  and  $y$  axes. Label the  $x$  axis with the midpoint of each class, and then use a suitable scale on the  $y$  axis for the frequencies.
- Step 3** Using the midpoints for the  $x$  values and the frequencies as the  $y$  values, plot the points.
- Step 4** Connect adjacent points with line segments. Draw a line back to the  $x$  axis at the beginning and end of the graph, at the same distance that the previous and next midpoints would be located, as shown in Figure 2-2.



# From the book

## The Ogive

The third type of graph that can be used represents the cumulative frequencies for the classes. This type of graph is called the *cumulative frequency graph*, or *ogive*. The **cumulative frequency** is the sum of the frequencies accumulated up to the upper boundary of a class in the distribution.

The **ogive** is a graph that represents the cumulative frequencies for the classes in a frequency distribution.

### EXAMPLE 2-6 Record High Temperatures

Construct an ogive for the frequency distribution described in Example 2-4.

#### SOLUTION

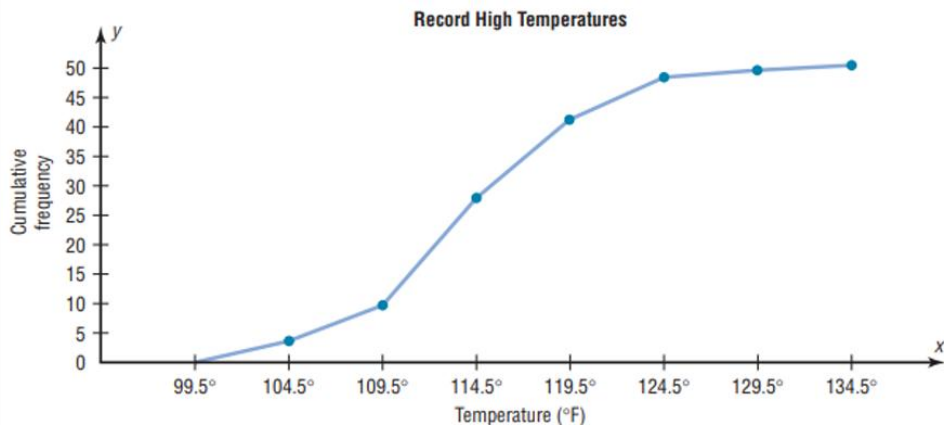
**Step 1** Find the cumulative frequency for each class.

|                 | Cumulative frequency |
|-----------------|----------------------|
| Less than 99.5  | 0                    |
| Less than 104.5 | 2                    |
| Less than 109.5 | 10                   |
| Less than 114.5 | 28                   |
| Less than 119.5 | 41                   |
| Less than 124.5 | 48                   |
| Less than 129.5 | 49                   |
| Less than 134.5 | 50                   |

**Step 2** Draw the  $x$  and  $y$  axes. Label the  $x$  axis with the class boundaries. Use an appropriate scale for the  $y$  axis to represent the cumulative frequencies. (Depending on the numbers in the cumulative frequency columns, scales such as 0, 1, 2, 3, . . . , or 5, 10, 15, 20, . . . , or 1000, 2000, 3000, . . . can be used. Do *not* label the  $y$  axis with the numbers in the cumulative frequency column.) In this example, a scale of 0, 5, 10, 15, . . . will be used.

**Step 3** Plot the cumulative frequency at each upper class boundary, as shown in Figure 2-3. Upper boundaries are used since the cumulative frequencies represent the number of data values accumulated up to the upper boundary of each class.

**Step 4** Starting with the first upper class boundary, 104.5, connect adjacent points with line segments, as shown in Figure 2-4. Then extend the graph to the first lower class boundary, 99.5, on the  $x$  axis.







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# Thank You

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